

Deep Learning Approaches for Identifying Metabolic Vulnerabilities in Breast Cancer Using Spatial Transcriptomics

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Abstract

Metabolic reprogramming is a hallmark of cancer, but the spatial heterogeneity of metabolic programmes within the tumour microenvironment remains poorly characterised. Spatially resolved and cross-patient conserved metabolic vulnerabilities are needed to inform targeted therapies. We present a spatially aware deep-learning framework for identifying proliferation-linked metabolic vulnerabilities in breast cancer using 10x Genomics Visium spatial transcriptomics. We analysed 24,422 spatial spots from seven breast cancer patients and included cross-platform evaluation using Visium HD (16 μm resolution). Our framework combines phenotype-linked prediction of proliferation from pathway activity, label-independent vulnerability scoring using autoencoder perturbation, and mechanistic support from flux inference with scFEA. In leave-one-patient-out validation, the classifier achieved robust performance (mean AUC = 0.8813 ± 0.0604) and generalised to the Visium HD sample (AUC = 0.8275), supporting cross-platform applicability. Flux analysis identified seven metabolic modules that were consistently associated with proliferation across all patients. Nucleotide metabolism emerged as the most reproducible vulnerability, with purine salvage and biosynthesis (M_{148}), pyrimidine biosynthesis (M_{155}), and nucleotide salvage (M_{159}) ranking as the highest-confidence targets across both Visium and Visium HD. Autoencoder-derived vulnerability scores showed significant correlation with proliferation ($r = 0.30$, $p < 10^{-100}$), indicating that metabolically essential programmes are enriched in proliferative regions. Together, these results provide a reproducible template for spatial metabolic vulnerability discovery and prioritise nucleotide biosynthesis as a robust, clinically actionable dependency in proliferative breast cancer.

Keywords:

1. Introduction

Breast cancer remains a major cause of cancer morbidity and mortality worldwide. One reason it is difficult to treat is the extent of heterogeneity within tumours. Malignant clones coexist with stromal and immune compartments, and local microenvironmental pressures such as hypoxia, acidosis and nutrient limitation vary

across the tissue [1]. A central component of this heterogeneity is metabolic reprogramming. By rewiring metabolism, tumour cells can meet biosynthetic demands, maintain redox balance and adapt to changing local resources. Over the past decade, altered metabolism has increasingly been viewed not only as a consequence of oncogenic transformation, but also as a source of therapeutic opportunity. In this view, “metabolic vulnerabilities” arise when specific pathways, enzymes or transport processes become limiting in particular tumour contexts [1, 2, 3]. These vulnerabilities are often conditional, shaped by lineage, genomic background, niche-specific microenvironments and treatment history [1, 2]. Because the same pathway can play different roles in different regions of the same tumour, spatially resolved mapping of metabolic programmes is essential for translating these insights into actionable breast cancer therapies.

In breast cancer, many subtype and niche-specific metabolic dependencies have been described. Triple-negative breast cancer (TNBC), for example, has recurrent dependencies involving lipid and amino acid metabolism, antioxidant systems and mitochondrial plasticity, often linked to stem-like states and treatment resistance [4]. Work on ferroptosis, an iron-dependent form of regulated cell death driven by lipid peroxidation, has further highlighted vulnerabilities connected to iron handling, GPX4 and selenium biology, polyunsaturated lipid pools and glutathione metabolism [5]. Cancer stem-like populations, which contribute to recurrence and drug resistance, can switch between bioenergetic states and reprogramme glycolysis, oxidative phosphorylation, one-carbon metabolism and lipid metabolism in ways that differ from bulk tumour cells [6]. Even well known features such as aerobic glycolysis (the Warburg effect) vary substantially between patients and across tumour regions, which cautions against one-size-fits-all assumptions and reinforces the need for spatial analysis [7]. Together, these observations motivate approaches that can identify where metabolic programmes occur, which cell states carry them, and how they relate to microenvironmental structure and clinically relevant phenotypes.

Spatial transcriptomics (ST) is well suited to these questions because it preserves tissue architecture while measuring gene expression. This makes it possible to study how malignant, stromal, endothelial and immune compartments are organised, and how local interactions shape molecular states. Recent studies show that ST can dissect tumour niches, characterise microenvironmental crosstalk, and identify spatial programmes linked to therapy response and resistance [8]. At the same time, there are important methodological challenges, including spot-level cell mixing, resolution limits that vary by platform, artefacts introduced during sample handling, and batch or context effects [9]. The emerging consensus is that robust discovery of metabolic vulnerabilities from ST should model spatial structure explicitly, remain aware of localised cell states despite spot mixing, integrate across samples to support generalisation, and use pathway-level readouts that map high-dimensional expression to interpretable metabolic programmes.

Deep learning has also reshaped spatial transcriptomics analysis. ST data can naturally be represented as a graph, where spots or cells are nodes and spatial proximity defines edges. This has driven wide use of graph neural networks and

related architectures for clustering, spatial domain detection, denoising and integration [10, 11, 12, 13]. In parallel, histology-guided models can use H&E images to predict or refine gene expression patterns and recover biologically meaningful programmes in breast cancer [11, 14]. These developments show that incorporating spatial structure and morphology can stabilise estimates of local programmes and support cross-sample alignment. Related work in spatial and bulk microRNA analysis also emphasises that metabolic phenotypes often reflect coordinated regulatory networks rather than single genes, which strengthens the case for pathway- and network-aware modelling [15, 16].

Despite this progress, identifying metabolic vulnerabilities directly from spatial transcriptomics remains difficult. Pathway activity is not a simple linear function of mRNA abundance, but emerges from enzyme kinetics, isoform compensation, substrate and cofactor availability, and feedback regulation. Spot mixing in platforms such as 10x Genomics Visium further complicates assignment of pathway activity to specific cell types, yet cell-type context is often critical for interpreting druggability. In addition, clinically useful insights require more than descriptive maps. They require features that predict relevant outcomes such as subtype, proliferation, risk or therapy response, and that can be linked to metabolism in a mechanistically plausible way. Perspective and benchmarking work therefore highlights the importance of careful validation, robust domain calling, and reproducible integration pipelines [9, 17]. In breast cancer, disease-focused spatial studies demonstrate the potential of deep learning to refine tumour microenvironment structure and highlight microenvironments that may shape metabolic trade-offs and treatment sensitivity [8].

Bringing the metabolism and spatial transcriptomics literatures together suggest a practical strategy for vulnerability discovery. Current frameworks emphasise the need to identify cancer-specific metabolic dependencies in context, and to map vulnerabilities to the cellular compartments and niches where they matter most [1, 2, 3]. In TNBC, recurring candidates include lipid metabolism, redox buffering and anabolic programmes linked to stem-like traits [4]. Ferroptosis sensitivity depends on iron handling, GPX4 activity and lipid composition [18], and cancer stem-like states introduce further liabilities involving one-carbon metabolism, shifts between fatty-acid synthesis and oxidation, and mitochondrial function [6]. The balance between glycolytic and oxidative metabolism also varies across tumours [7], supporting spatially resolved, pathway-level inference rather than reliance on bulk averages.

In this study, we present a reproducible framework for spatially resolved metabolic vulnerability discovery in breast cancer. First, we introduce a spatially aware deep-learning pipeline, *MetabolicVulnerabilityNet*, and reduce leakage by using spatial block-wise cross-validation rather than random spot splits. Second, we implement a data-driven pathway discovery procedure that scores 7,829 curated gene sets across five resources (MSigDB Hallmark, KEGG, Reactome, GO Biological Process and BioCarta), applies coverage-based filtering, and controls multiple testing using Benjamini–Hochberg false-discovery rate correction. Third, we evaluate generalisation across seven breast cancer samples, including cross-platform assessment with Visium HD, using leave-one-patient-out validation. Fourth, we

integrate expression-derived pathway scores with reaction-level flux estimates from single-cell Flux Estimation Analysis (scFEA) across all patients to provide mechanistic support for transcriptional signals. Fifth, we benchmark against a regularised logistic regression baseline and include negative controls based on label and feature permutation to verify that performance reflects biological structure rather than artefacts. Together, these components provide a practical template for mapping and prioritising metabolic vulnerabilities in a spatial context.

Building on this framework, we define a spot-level proliferation target from canonical cell-cycle markers, perform stringent quality control, and evaluate models using spatially aware cross-validation that respects tissue architecture. We compute pathway activity scores by aggregating curated gene sets into spot-level measures, test pathway–proliferation associations under explicit false-discovery control, and train both regularised logistic regression and a compact neural network on the selected pathway features. To move beyond association, we perform pathway ablation and permutation analyses and quantify spatial coherence using autocorrelation statistics. We also train an autoencoder on pathway activity profiles to estimate metabolic essentiality through in silico knockdown, which provides a label-independent measure of pathway vulnerability. Finally, we integrate scFEA-derived flux modules across all patients to link expression-level signals to inferred metabolic activity and prioritise vulnerabilities supported by convergent evidence from expression and flux. Throughout, we follow best-practice guidance on quality control, leakage avoidance, domain awareness and reproducible analysis in spatial transcriptomics [9].

2. Methods

2.1. Dataset and study design

We analysed spatial transcriptomics data from seven breast cancer patients obtained using the 10x Genomics Visium platform. Six patients (Patients 1–6) were profiled using standard Visium (55 μm spot diameter), whilst Patient 7 was profiled using the higher-resolution Visium HD platform (16 μm bin size). The dataset comprised approximately 4,869, 2,516, 3,988, 1,642, 4,166, 1,642, and 5,599 spots for Patients 1–7, respectively (total: 24,422 spots; average: 3,489 spots per patient). Each standard Visium spot captures expression from roughly 1–10 cells, whilst Visium HD bins provide near-single-cell resolution. Gene expression was measured using standard 10x Genomics protocols and aligned to the human reference genome (GRCh38) according to the manufacturer’s recommendations.

Patient 1 served as the primary discovery cohort for model development and internal validation. Patients 2–6 represent additional breast cancer samples with matched Visium spatial profiles used for leave-one-patient-out (LOPO) cross-validation. Patient 7 (Visium HD) provided cross-platform validation to assess whether findings generalise across different spatial transcriptomics technologies. For all external patients, we applied the same quality-control (QC) and feature-construction pipeline as for the primary cohort. All analysis code and preprocessing pipelines used in this study are publicly available at <https://github.com/hoseaisaac/Metabolic-Vulnerabilities-in-Breast-Cancer-Using-Spatial-Transcriptomics>.

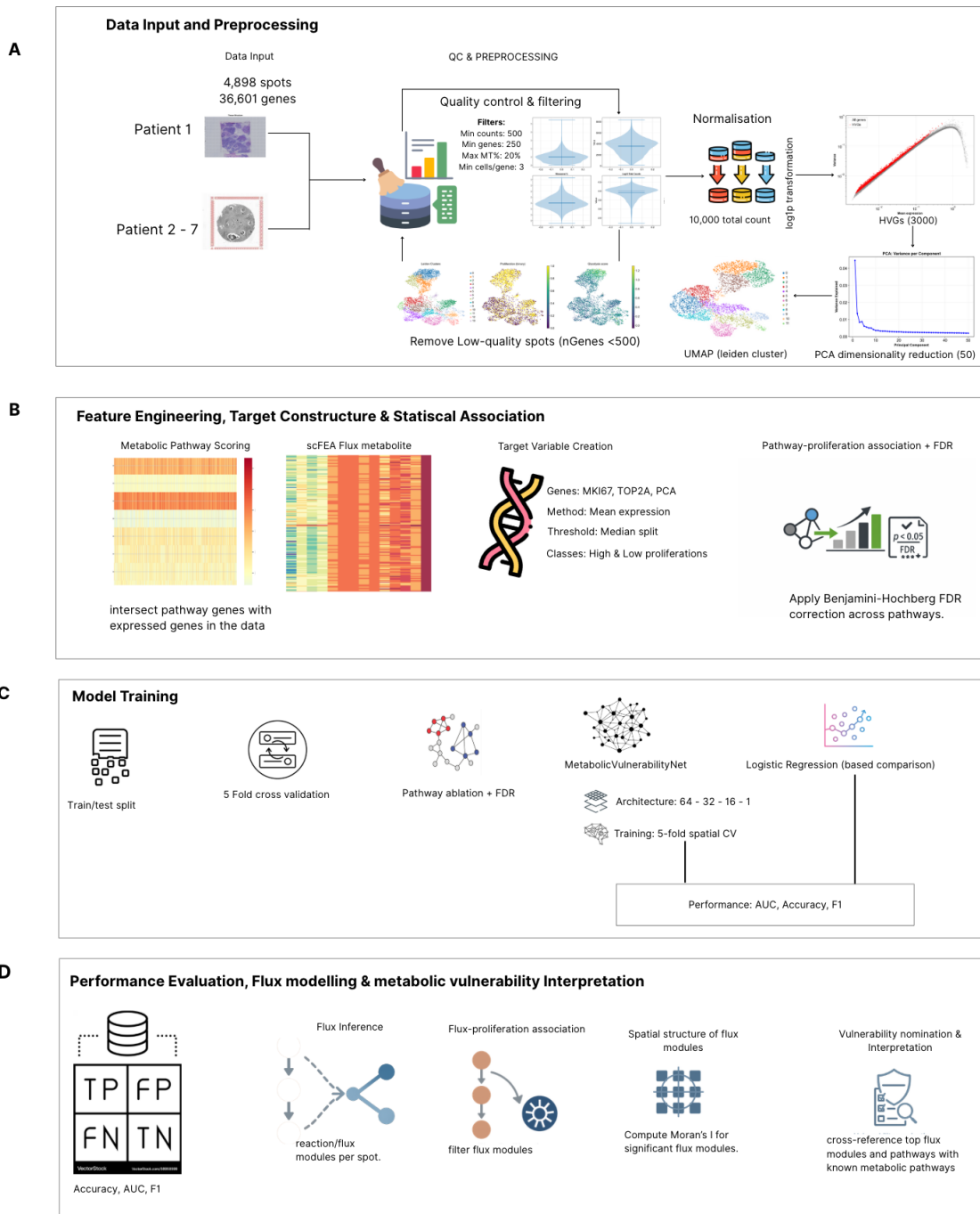


Figure 1: Overview of the analysis pipeline for identifying metabolic vulnerabilities in breast cancer using spatial transcriptomics. (A) Data input and preprocessing: Visium breast tumour sections from seven patients undergo spot- and gene-level quality control, normalisation, and dimensionality reduction (PCA/UMAP). (B) Pathway and label construction: pathway activity scores are computed from 7,829 gene sets across five databases, a binary proliferation target is defined from canonical cell-cycle markers, and pathway–proliferation associations are tested with Benjamini–Hochberg FDR correction. (C) Model training and internal validation: MetabolicVulnerabilityNet is trained with 5-fold spatial block-wise cross-validation and benchmarked against logistic regression; performance metrics (AUC-ROC, accuracy, F1), feature importance, and negative controls (label/pathway permutation) are evaluated. (D) External validation: model generalisability is assessed via leave-one-patient-out cross-validation across seven patients, including cross-platform validation with Visium HD. (E) Flux modelling and target nomination: reaction-level fluxes are inferred using scFEA for all patients, cross-patient comparison identifies universal metabolic modules, and therapeutic targets are prioritised based on convergent evidence from expression and flux analyses.

2.2. Pre-processing and quality control

All pre-processing was performed in Python using Scanpy (version 1.10.3). Let N denote the number of spots and G the number of genes for a given patient. We write X_{ig} for the raw count of gene $g \in \{1, \dots, G\}$ in spot $i \in \{1, \dots, N\}$, and collect these into a count matrix $\mathbf{X} \in \mathbb{N}_0^{N \times G}$.

2.2.1. QC metric calculation

For each spot i , we computed:

$$n_i = \sum_{g=1}^G X_{ig} \quad (\text{total counts per spot}), \quad (1)$$

$$k_i = |\{g : X_{ig} > 0\}| \quad (\text{number of detected genes}), \quad (2)$$

$$m_i = \frac{\sum_{g \in \mathcal{M}} X_{ig}}{\sum_{g=1}^G X_{ig}} \times 100 \quad (\text{percentage of mitochondrial counts}), \quad (3)$$

where $\mathcal{M}$ denotes the set of mitochondrial genes (symbols starting with MT-).

2.2.2. Spot and gene filtering

Spots were retained if they passed the following thresholds:

$$\mathbb{I}_i^{\text{spot}} = \mathbb{I}(n_i \geq n_{\min} \wedge k_i \geq k_{\min} \wedge m_i \leq m_{\max}), \quad (4)$$

with $n_{\min} = 500$, $k_{\min} = 200$, and $m_{\max} = 20$ (per cent). Genes were retained if

$$\mathbb{I}_g^{\text{gene}} = \mathbb{I}(|\{i : X_{ig} > 0\}| \geq c_{\min}), \quad (5)$$

with $c_{\min} = 5$. Only spots and genes with indicator equal to one were kept.

2.2.3. Visium HD preprocessing

For Patient 7 (Visium HD), the raw data contained 167,545 spots at 16 μm resolution. To ensure computational tractability and comparability with standard Visium data, we applied random downsampling to 15,000 spots prior to QC filtering, using a fixed random seed (42) for reproducibility. After QC, 5,599 spots were retained for downstream analysis.

2.2.4. Normalisation and log-transformation

For each spot i , we performed library-size normalisation to a fixed depth of 10,000 counts:

$$\tilde{X}_{ig} = \frac{10^4 \cdot X_{ig}}{\sum_{g'=1}^G X_{ig'}}. \quad (6)$$

We then applied a log-transform with a pseudocount of one:

$$Z_{ig} = \log(1 + \tilde{X}_{ig}), \quad (7)$$

yielding a matrix $\mathbf{Z} \in \mathbb{R}^{N \times G}$ used for downstream analyses.

2.2.5. Highly variable genes and dimensionality reduction

Highly variable genes (HVGs) were selected using Scanpy’s variance-stabilisation procedure (flavour: `cell_ranger`), retaining the top 3,000 genes. Principal component analysis (PCA) was performed on the HVG subset, and we retained $d = 30$ principal components based on the elbow of the variance-explained curve. Let $\mathbf{P} \in \mathbb{R}^{N \times d}$ denote the matrix of PC scores.

2.2.6. Neighbour graph, UMAP, and clustering

A k -nearest neighbour graph was constructed on the rows of $\mathbf{P}$ (with $k = 15$). On this graph, we computed a two-dimensional UMAP embedding (minimum distance = 0.3, spread = 1.0) and applied Leiden clustering (resolution = 0.5) for unsupervised identification of spatial domains, yielding 15 clusters for Patient 1. These representations were used primarily for visual QC and spatial domain characterisation.

For reproducibility, random seeds were fixed at 42 for all stochastic steps (PCA initialisation, neighbour graph construction, UMAP, and clustering).

2.3. Proliferation target definition

Our supervised models were trained to predict a binary proliferation label derived from a canonical proliferation gene signature. Let $\mathcal{G}_{\text{prolif}}$ denote the set of proliferation genes (*MKI67*, *TOP2A*, *PCNA*, *CCNB1*, *CCNB2*, *CDK1*, *BIRC5*) that are present in the dataset.

For each spot i , we defined a continuous proliferation score:

$$s_i^{\text{prolif}} = \frac{1}{|\mathcal{G}_{\text{prolif}}|} \sum_{g \in \mathcal{G}_{\text{prolif}}} Z_{ig}. \quad (8)$$

For the primary cohort (Patient 1), this score was binarised at the 75th percentile:

$$y_i = \mathbb{I} \left(s_i^{\text{prolif}} > Q_{0.75}(s^{\text{prolif}}) \right), \quad y_i \in \{0, 1\}, \quad (9)$$

where $Q_{0.75}(\cdot)$ denotes the 75th percentile, classifying the top 25% of spots as “high proliferation” ($y_i = 1$). For cross-validation within Patient 1, the threshold was adjusted to achieve 50% class balance (median threshold) to facilitate model training. For external patients, the same percentile-based threshold (75th percentile) was applied independently to each sample, yielding class proportions ranging from 5.6% (Patient 7, Visium HD) to 50% (Patient 1).

2.4. Pathway annotation and scoring

To move from individual genes to interpretable metabolic programmes, we aggregated genes into curated pathways drawn from multiple sources: MSigDB Hallmark 2020 (50 pathways), KEGG 2021 Human (320 pathways), Reactome 2022 (1,816 pathways), GO Biological Process 2023 (5,406 pathways), and BioCarta 2016 (237 pathways), yielding a total of 7,829 candidate pathways accessed via the GSEAPy library. Let $\mathcal{P}$ denote the set of pathways and, for each pathway $p \in \mathcal{P}$, let $\mathcal{G}_p$ be the set of constituent genes.

After intersecting with the genes present in the Visium dataset, pathways with fewer than three overlapping genes ($|\mathcal{G}_p| < 3$) or less than 20% gene coverage were discarded. For each retained pathway p and spot i , we defined the pathway activity score as:

$$s_{ip}^{\text{path}} = \frac{1}{|\mathcal{G}_p|} \sum_{g \in \mathcal{G}_p} Z_{ig}. \quad (10)$$

Collecting these scores over all spots yields a matrix:

$$\mathbf{S}^{\text{path}} = \left(s_{ip}^{\text{path}} \right)_{i=1, \dots, N; p \in \mathcal{P}} \in \mathbb{R}^{N \times |\mathcal{P}|}, \quad (11)$$

which we stored alongside the spot-level metadata. After filtering, 2,427 pathways were retained for Patient 1, with similar numbers (2,066–2,154) retained across external patients.

2.5. Technical covariates

To account for technical variation, we computed spot-level covariates from the count matrix:

$$t_i^{\text{total}} = \log(n_i + 1), \quad (12)$$

$$t_i^{\text{mt}} = m_i, \quad (13)$$

$$t_i^{\text{ribo}} = \frac{\sum_{g \in \mathcal{R}} X_{ig}}{\sum_{g=1}^G X_{ig}} \times 100, \quad (14)$$

where $\mathcal{R}$ is the set of ribosomal protein genes (symbols starting with RPS or RPL; 97 genes detected). These three covariates were z -score standardised and concatenated with pathway scores to form the feature vector $\mathbf{x}_i$ for each spot.

2.6. Selection of candidate metabolic pathways

We focused on pathways likely to encode actionable metabolic vulnerabilities. For each pathway $p \in \mathcal{P}$, we quantified the association between the pathway score and the proliferation label using Spearman’s rank correlation:

$$\rho_p = \text{CORR}_{\text{Spearman}} \left(s_{1p}^{\text{path}}, \dots, s_{Np}^{\text{path}}; y_1, \dots, y_N \right), \quad (15)$$

and a Mann–Whitney U test comparing $\{s_{ip}^{\text{path}} : y_i = 1\}$ and $\{s_{ip}^{\text{path}} : y_i = 0\}$.

The resulting p -values p_p were adjusted using the Benjamini–Hochberg procedure to yield q_p . Pathways with $q_p < 0.05$ and $|\rho_p| > 0.1$ were considered significantly associated with proliferation. The top 20 pathways ranked by $|\rho_p|$, together with the three technical covariates, formed the primary feature set ($d = 23$ features) for model training. The selected pathways included cell-cycle regulators (e.g., REACTOM E_G2_M_DNA_REPLICATION_CHECKPOINT and KEGG_MEDICUS_REFERENCE_WEE1_CELL_CYCLE_G2_M) and DNA damage response pathways (e.g., BIOCARTA_DNAFRAGMENT_PATHWAY).

2.7. Spatial cross-validation framework

To respect spatial structure and prevent data leakage from spatially autocorrelated observations, we implemented a spatial block-wise cross-validation strategy. Spots were partitioned into $K = 5$ spatial blocks using k -means clustering on their physical coordinates (x_i, y_i) :

$$g_i = \arg \min_{c \in \{1, \dots, K\}} \|(x_i, y_i) - \boldsymbol{\mu}_c\|_2, \quad (16)$$

where $\boldsymbol{\mu}_c$ denotes the centroid of block c . Block sizes ranged from 842 to 1,160 spots (17.3%–23.8% of total). We then applied GroupKFold cross-validation, successively holding out each spatial block as a test set and training on the remaining blocks. This ensures that spatially proximal (and thus potentially correlated) spots are never split between training and validation sets.

2.8. Predictive modelling of proliferation

Let $\mathbf{x}_i \in \mathbb{R}^d$ denote the feature vector for spot i , obtained by concatenating the 20 selected pathway scores and 3 technical covariates (after z -score standardisation), and let $y_i \in \{0, 1\}$ denote the proliferation label.

2.8.1. Baseline: regularised logistic regression

As a baseline model, we used ℓ_2 -regularised logistic regression. The model assumes:

$$\Pr(y_i = 1 \mid \mathbf{x}_i) = \sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (17)$$

where $\beta_0 \in \mathbb{R}$ is an intercept and $\boldsymbol{\beta} \in \mathbb{R}^d$ is a vector of coefficients. Parameters were estimated by minimising the regularised binary cross-entropy loss:

$$\mathcal{L}(\beta_0, \boldsymbol{\beta}) = -\frac{1}{N} \sum_{i=1}^N [y_i \log p_i + (1 - y_i) \log(1 - p_i)] + \lambda \|\boldsymbol{\beta}\|_2^2, \quad (18)$$

where $p_i = \Pr(y_i = 1 \mid \mathbf{x}_i)$ and $\lambda = 0.01$ is the regularisation strength. Under 5-fold spatial cross-validation, the baseline achieved mean AUC = 0.8577 ± 0.0275 .

2.8.2. Deep neural network: MetabolicVulnerabilityNet

To capture potential non-linear interactions among pathways, we implemented a feedforward neural network, *MetabolicVulnerabilityNet* (version 1.0.0), defined as a function:

$$f_{\boldsymbol{\theta}} : \mathbb{R}^d \rightarrow (0, 1), \quad (19)$$

where $\boldsymbol{\theta}$ collects all network parameters (6,113 trainable parameters). The architecture consisted of three hidden layers with 64, 32, and 16 units respectively, each

followed by batch normalisation, rectified linear unit (ReLU) activation, and dropout (rate = 0.3):

$$\mathbf{h}_1 = \text{Dropout}(\text{ReLU}(\text{BN}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1))), p = 0.3), \quad (20)$$

$$\mathbf{h}_2 = \text{Dropout}(\text{ReLU}(\text{BN}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2))), p = 0.3), \quad (21)$$

$$\mathbf{h}_3 = \text{Dropout}(\text{ReLU}(\text{BN}(\mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3))), p = 0.3), \quad (22)$$

$$p_i = \sigma(\mathbf{w}_{\text{out}}^\top \mathbf{h}_3 + b_{\text{out}}), \quad (23)$$

where $\mathbf{W}_1 \in \mathbb{R}^{64 \times d}$, $\mathbf{W}_2 \in \mathbb{R}^{32 \times 64}$, $\mathbf{W}_3 \in \mathbb{R}^{16 \times 32}$, $\mathbf{w}_{\text{out}} \in \mathbb{R}^{16}$, and BN denotes batch normalisation.

Parameters were learnt by minimising the binary cross-entropy loss:

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{i=1}^N [y_i \log p_i + (1 - y_i) \log(1 - p_i)], \quad (24)$$

using the Adam optimiser with learning rate $\eta = 10^{-3}$, weight decay 10^{-2} , and batch size 64. Training proceeded for a maximum of 150 epochs with early stopping (patience = 20 epochs) based on validation AUC. Under 5-fold spatial cross-validation, the neural network achieved mean AUC = 0.8661 ± 0.0260 , a modest improvement over the logistic regression baseline.

2.9. Model calibration assessment

To evaluate whether predicted probabilities accurately reflect true outcome frequencies, we assessed model calibration using three complementary metrics.

2.9.1. Brier score

The Brier score measures the mean squared error between predicted probabilities and binary outcomes:

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2, \quad (25)$$

where BS = 0 indicates perfect calibration and BS = 0.25 corresponds to random prediction. The final model achieved BS = 0.1289.

2.9.2. Expected calibration error

We computed the expected calibration error (ECE) by partitioning predictions into $M = 10$ equal-frequency bins and measuring the weighted average of calibration gaps:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (26)$$

where B_m is the set of samples in bin m , $\text{acc}(B_m)$ is the empirical accuracy, and $\text{conf}(B_m)$ is the mean predicted probability within the bin. The model achieved ECE = 0.0254, indicating excellent calibration.

2.9.3. Precision-recall AUC

Given the potential for class imbalance across patients, we computed the area under the precision-recall curve (PR-AUC), which is more informative than ROC-AUC for imbalanced datasets. The model achieved PR-AUC = 0.9080, compared to a baseline of 0.50 for random prediction.

2.10. Autoencoder-based metabolic vulnerability analysis

To complement the supervised classification approach, we developed an unsupervised autoencoder framework to identify metabolic vulnerabilities through in-silico pathway perturbation. This approach quantifies how dependent the overall metabolic state of each spot is on individual pathways, independent of proliferation labels.

2.10.1. Autoencoder architecture

We trained a symmetric autoencoder on the pathway activity scores. Let $\mathbf{s}_i = (s_{i1}^{\text{path}}, \dots, s_{i|\mathcal{P}_{\text{sel}}|}^{\text{path}})^{\top} \in \mathbb{R}^P$ denote the vector of $P = 20$ selected pathway scores for spot i (excluding technical covariates). The autoencoder consists of an encoder $E_{\phi} : \mathbb{R}^P \rightarrow \mathbb{R}^L$ and a decoder $D_{\psi} : \mathbb{R}^L \rightarrow \mathbb{R}^P$, where L is the latent dimension.

The encoder compresses the input through two hidden layers:

$$\mathbf{e}_1 = \text{ReLU} \left(\mathbf{W}_1^{(E)} \mathbf{s}_i + \mathbf{b}_1^{(E)} \right), \quad \mathbf{W}_1^{(E)} \in \mathbb{R}^{32 \times P}, \quad (27)$$

$$\mathbf{z}_i = \text{ReLU} \left(\mathbf{W}_2^{(E)} \mathbf{e}_1 + \mathbf{b}_2^{(E)} \right), \quad \mathbf{W}_2^{(E)} \in \mathbb{R}^{L \times 32}, \quad (28)$$

where $L = 8$ is the bottleneck dimension. The decoder reconstructs the input symmetrically:

$$\mathbf{d}_1 = \text{ReLU} \left(\mathbf{W}_1^{(D)} \mathbf{z}_i + \mathbf{b}_1^{(D)} \right), \quad \mathbf{W}_1^{(D)} \in \mathbb{R}^{32 \times L}, \quad (29)$$

$$\hat{\mathbf{s}}_i = \mathbf{W}_2^{(D)} \mathbf{d}_1 + \mathbf{b}_2^{(D)}, \quad \mathbf{W}_2^{(D)} \in \mathbb{R}^{P \times 32}, \quad (30)$$

where $\hat{\mathbf{s}}_i$ is the reconstructed pathway score vector. The full architecture is thus $P \rightarrow 32 \rightarrow 8 \rightarrow 32 \rightarrow P$ (i.e., $20 \rightarrow 32 \rightarrow 8 \rightarrow 32 \rightarrow 20$).

2.10.2. Training objective

The autoencoder was trained to minimise the mean squared error (MSE) between input and reconstructed pathway scores:

$$\mathcal{L}_{\text{AE}}(\phi, \psi) = \frac{1}{N} \sum_{i=1}^N \|\mathbf{s}_i - \hat{\mathbf{s}}_i\|_2^2 = \frac{1}{N} \sum_{i=1}^N \sum_{p=1}^P \left(s_{ip}^{\text{path}} - \hat{s}_{ip} \right)^2. \quad (31)$$

Training was performed using the Adam optimiser (learning rate $\eta = 10^{-3}$, weight decay 10^{-5}) for 100 epochs with batch size 128. We applied an 80/20 train/validation split and used early stopping (patience = 15 epochs) based on validation MSE. Input features were z -score standardised using training set statistics.

2.10.3. In-silico pathway knockdown

To quantify the contribution of each pathway to the learned metabolic representation, we performed in-silico knockdown experiments. For each pathway $p \in \{1, \dots, P\}$ and each spot i , we created a perturbed input vector $\mathbf{s}_i^{(-p)}$ by setting the p -th pathway score to zero (simulating complete pathway ablation):

$$s_{ij}^{(-p)} = \begin{cases} 0 & \text{if } j = p, \\ s_{ij}^{\text{path}} & \text{if } j \neq p. \end{cases} \quad (32)$$

We chose zero-imputation rather than mean-imputation because it represents a more biologically interpretable “loss-of-function” scenario where pathway activity is completely ablated.

2.10.4. Reconstruction error and pathway vulnerability score

For each spot i and knocked-down pathway p , we computed the reconstruction error under perturbation:

$$\text{RE}_i^{(-p)} = \left\| \mathbf{s}_i - D_\psi \left(E_\phi \left(\mathbf{s}_i^{(-p)} \right) \right) \right\|_2^2, \quad (33)$$

where the reconstruction target remains the *original* (unperturbed) pathway scores $\mathbf{s}_i$. We also computed the baseline reconstruction error without perturbation:

$$\text{RE}_i^{(0)} = \left\| \mathbf{s}_i - \hat{\mathbf{s}}_i \right\|_2^2. \quad (34)$$

The change in reconstruction error due to knockdown of pathway p in spot i is:

$$\Delta \text{RE}_{ip} = \text{RE}_i^{(-p)} - \text{RE}_i^{(0)}. \quad (35)$$

A large positive ΔRE_{ip} indicates that pathway p is critical for reconstructing the metabolic state of spot i —i.e., the autoencoder has learnt that pathway p carries substantial information about the overall metabolic programme at that location.

2.10.5. Global pathway vulnerability ranking

To obtain a global vulnerability score for each pathway, we averaged the change in reconstruction error across all spots:

$$V_p^{\text{AE}} = \frac{1}{N} \sum_{i=1}^N \Delta \text{RE}_{ip}. \quad (36)$$

Pathways were ranked by V_p^{AE} , with higher values indicating greater metabolic essentiality. We also computed per-spot vulnerability profiles and visualised these spatially to identify regions where specific pathways are most critical.

2.10.6. Deep-learning vulnerability score

The *deep-learning vulnerability score* for each spot i was defined as the sum of reconstruction error changes across all pathway knockdowns:

$$V_i^{\text{DL}} = \sum_{p=1}^P \Delta \text{RE}_{ip}. \quad (37)$$

This score quantifies the overall metabolic fragility of each spot: spots with high V_i^{DL} have metabolic states that are highly sensitive to pathway perturbations and are thus predicted to harbour greater metabolic vulnerabilities. Importantly, this score is derived entirely from the *unsupervised autoencoder*, not from the supervised classifier (MetabolicVulnerabilityNet). The classifier predicts proliferation status, whilst the autoencoder-derived vulnerability score captures metabolic essentiality independent of the proliferation label.

To validate the biological relevance of V_i^{DL} , we computed its Pearson and Spearman correlation with the proliferation score s_i^{prolif} and visualised both metrics spatially to assess co-localisation of high vulnerability and high proliferation regions.

2.11. Pathway ablation analysis for the classifier

To quantify the contribution of individual pathways to the supervised classification task, we performed leave-one-out ablation experiments using the logistic regression model. Let $A^{(k)}$ denote the AUC obtained in fold k with the full feature set, and let $A_{-p}^{(k)}$ be the AUC when the feature corresponding to pathway p is removed from $\mathbf{x}_i$.

For each pathway p and fold $k \in \{1, \dots, K\}$, we defined the fold-specific performance change:

$$\Delta_p^{(k)} = A^{(k)} - A_{-p}^{(k)}. \quad (38)$$

A positive Δ_p indicates that removing pathway p *decreases* performance, suggesting the pathway contributes positively to prediction. We summarised effects by the mean change:

$$\Delta_p = \frac{1}{K} \sum_{k=1}^K \Delta_p^{(k)}, \quad (39)$$

and constructed 95% confidence intervals via bootstrap resampling (1,000 iterations).

The top-ranked pathways by ablation effect were *BIOCARTA_DNAFRAGMENT_PATHWAY* ($\Delta = 0.0233$) and *REACTOME_TRANSCRIPTION_OF_E2F_TARGETS* ($\Delta = 0.0207$). After Benjamini–Hochberg correction, no individual pathway survived $\text{FDR} < 0.05$, consistent with the distributed nature of metabolic information across multiple pathways.

2.12. Permutation importance analysis

To complement ablation analysis, we computed permutation feature importance for the trained neural network. For each feature j , we randomly shuffled its values

across all spots (30 repeats) and measured the decrease in AUC:

$$I_j = \text{AUC}_{\text{original}} - \frac{1}{R} \sum_{r=1}^R \text{AUC}_{j,r}^{\text{permuted}}, \quad (40)$$

where $R = 30$ is the number of permutation repeats. Features with high I_j are those for which the model’s predictions are most sensitive to perturbation. The top features by permutation importance were *REACTOME_TRANSCRIPTION_OF_E2F_TARGETS* ($I = 0.0293$), *BIOCARTA_DNAFRAGMENT_PATHWAY* ($I = 0.0285$), and the technical covariate *n_counts* ($I = 0.0203$).

2.13. Negative control experiments

To validate that model performance reflects genuine biological signal rather than technical artefacts or overfitting, we implemented two negative control analyses.

2.13.1. Label permutation control

We randomly shuffled the proliferation labels $\{y_i\}_{i=1}^N$ whilst keeping the feature matrix $\mathbf{X}$ fixed, thereby breaking any true association between features and labels. We repeated this procedure $B = 100$ times, retraining the classifier and recording the AUC for each permutation. Under the null hypothesis of no association, the AUC distribution was centred at 0.5. We computed an empirical permutation p -value as:

$$p_{\text{perm}} = \frac{1 + \sum_{b=1}^B \mathbb{I}(\text{AUC}_b \geq \text{AUC}_{\text{obs}})}{1 + B}, \quad (41)$$

where AUC_{obs} is the observed AUC from the original data. The observed AUC (0.8577) exceeded all 100 permuted values, yielding an empirical permutation p -value < 0.01 .

2.13.2. Pathway score permutation control

To test whether the spatial organisation of pathway activity contributes to predictive performance, we permuted individual pathway scores across spots whilst preserving the marginal distribution of each pathway. For all 20 pathways, the real ablation effect significantly exceeded the permuted null distribution (all $p < 0.05$), indicating that the spatial coherence of pathway activity not merely the presence of certain expression levels is informative for proliferation prediction.

2.14. Leave-one-patient-out cross-validation

To test whether the identified metabolic signatures generalise across patients, we performed leave-one-patient-out (LOPO) cross-validation using all seven samples. Let $\mathcal{I}_j$ denote the index set of spots belonging to Patient $j \in \{1, \dots, 7\}$. In each LOPO fold, Patient j was held out as the test set and the remaining six patients formed the training set $\bigcup_{k \neq j} \mathcal{I}_k$. The same quality control, normalisation and pathway-scoring pipeline was applied to each patient to construct comparable

features, and feature standardisation was carried out using z -score parameters estimated from the training patients only to avoid information leakage. To ensure consistent feature alignment across folds, pathway gene sets were defined from the Patient 1 feature definitions and then applied unchanged to all patients. A *Metabolic VulnerabilityNet* with an identical architecture was trained on the training set for 30 epochs, and performance on the held-out patient was summarised using AUC, accuracy, sensitivity, specificity and F1 score.

Results were summarised by mean and standard deviation across the seven LOPO folds. The model achieved mean AUC = 0.8813 ± 0.0604 , mean accuracy = 0.8439 ± 0.0696 , and mean F1 = 0.6194 ± 0.1949 . Individual patient AUCs ranged from 0.8041 (Patient 3) to 0.9440 (Patient 5), demonstrating robust cross-patient generalisation. Notably, Patient 7 (Visium HD) achieved AUC = 0.8275, validating cross-platform applicability.

2.15. Flux inference with scFEA

To complement expression-based pathway scores with estimates of metabolic flux, we integrated our data with the single-cell Flux Estimation Analysis (scFEA) framework. scFEA infers reaction-level metabolic fluxes from gene expression data using a graph neural network trained on a human metabolic network comprising 168 metabolic modules.

For each patient, we exported the filtered, non-normalised count matrix $\mathbf{X}$ in gene-by-spot orientation as required by scFEA. To ensure computational tractability, large datasets were subsampled: Patient 7 (Visium HD) was downsampled to 5,000 spots prior to scFEA execution. scFEA was run with default parameters and 50 training epochs, yielding estimated fluxes F_{im} for each spot i and metabolic module m . After filtering modules with near-zero variance, the number of informative flux modules ranged from 73 (Patient 5) to 120 (Patient 7) across patients.

2.16. Flux-proliferation association analysis

For each patient independently, we quantified the association between each flux module and proliferation status using Spearman correlation:

$$\rho_m^{\text{flux}} = \text{corr}_{\text{Spearman}}(F_{1m}, \dots, F_{Nm}; y_1, \dots, y_N). \quad (42)$$

Mann–Whitney U tests were performed comparing flux values between high and low proliferation spots, with Benjamini–Hochberg FDR correction applied per patient. A module was considered significantly associated with proliferation if $q < 0.05$. The number of significant modules varied across patients: Patient 1 (69/79), Patient 2 (97/103), Patient 3 (79/96), Patient 4 (80/83), Patient 5 (73/73), Patient 6 (88/90), and Patient 7 (54/120).

2.17. Cross-patient flux comparison

To identify metabolic vulnerabilities conserved across the cohort, we compared flux–proliferation associations across patients. For each flux module m observed in at least four patients, let P_m denote the set of patients with available estimates for that

module. We summarised the strength and consistency of association by computing the mean Spearman correlation across patients,

$$\bar{\rho}_m = \frac{1}{|P_m|} \sum_{j \in P_m} \rho_{m,j}, \quad (43)$$

and the number of patients in which the module was significantly associated with proliferation after multiple-testing correction,

$$n_m^{\text{sig}} = \sum_{j \in P_m} \mathbb{I}(q_{m,j} < 0.05). \quad (44)$$

We then defined a conservation criterion to identify broadly reproducible modules, labelling a module as “conserved” if it was significant in at least half of the patients with data for that module, that is if $n_m^{\text{sig}} \geq |P_m|/2$.

Across 158 unique modules, 76 were conserved (significant in $\geq 50\%$ of patients), and 7 were “universal” (significant in all 7 patients).

2.18. Cross-platform validation

To assess whether findings generalise across different Visium technologies, we compared flux–proliferation associations between standard Visium (Patients 1–6) and Visium HD (Patient 7). For each module, we computed:

$$\bar{\rho}_m^{\text{standard}} = \frac{1}{6} \sum_{j=1}^6 \rho_{m,j}, \quad (45)$$

$$\rho_m^{\text{HD}} = \rho_{m,7}. \quad (46)$$

Modules were considered cross-platform validated if they were significant in at least one standard Visium patient AND in Visium HD, with consistent direction of effect. Twenty-five modules met these criteria, including the top-ranked universal targets (M_148, M_155, M_159).

2.19. Metabolic module annotation

To interpret flux modules biologically, we mapped scFEA module identifiers to metabolic pathways using the module–gene annotation file (`module_gene_m168.csv`). For each module, we extracted the associated genes and assigned a putative metabolic process based on known gene functions and pathway membership. Among the universally significant modules, M_148 was annotated as purine salvage and biosynthesis and included genes such as *APRT*, *GMPS* and *HPRT1*, which are clinically targetable using antifolate and purine analogue therapies (for example methotrexate and 6-mercaptopurine). M_155 corresponded to pyrimidine biosynthesis and contained *CTPS1* and *CTPS2*, consistent with vulnerability to agents that disrupt nucleotide synthesis (for example gemcitabine). M_157 captured ribonucleotide reductase activity through *RRM1*, *RRM2* and *RRM2B*, supporting sensitivity to ribonucleotide reductase inhibitors such as hydroxyurea or triapine

and to nucleoside analogues including gemcitabine. In addition, M_2 mapped to glycolysis and included genes such as *ALDOA*, *PFKL* and *GPI*, which are commonly discussed in the context of glycolytic inhibition (for example 2-deoxyglucose and lonidamine). Finally, M_8 was annotated as the tricarboxylic acid (TCA) cycle and included *ACO1*, *IDH1* and *IDH2*, linking this module to therapeutic strategies that target mitochondrial metabolism (for example CPI-613 and IDH-directed agents such as ivosidenib).

2.20. Expression-flux integration

To jointly interpret expression-derived pathway activity and scFEA-inferred flux, we quantified convergent evidence from both modalities. For biologically matched pathway-module pairs (e.g., glycolysis pathway score and glycolysis flux module), we computed an integrated vulnerability score:

$$V_{p,m}^{\text{int}} = |\rho_p^{\text{expr}}| \times |\rho_m^{\text{flux}}|, \quad (47)$$

where ρ_p^{expr} is the expression-based pathway correlation and ρ_m^{flux} is the flux-based module correlation. Pairs with high $V_{p,m}^{\text{int}}$ and significant associations in both modalities represent high-confidence metabolic vulnerabilities with convergent transcriptional and metabolic evidence.

2.21. Spatial metabolic niche identification

To characterise the spatial organisation of metabolic programmes within the tumour microenvironment, we performed unsupervised clustering of pathway activity scores. For Patient 1, we applied Leiden clustering (resolution = 0.5) on the pathway score matrix $\mathbf{S}^{\text{path}}$, yielding 15 distinct metabolic clusters. Cluster assignments were mapped back to spatial coordinates to visualise metabolic niches within the tissue architecture.

For each cluster c , we computed the proportion of high-proliferation spots:

$$\pi_c^{\text{prolif}} = \frac{\sum_{i \in c} y_i}{|c|}, \quad (48)$$

and tested for significant enrichment using a χ^2 test comparing observed versus expected frequencies under the null hypothesis of uniform distribution. Clusters with $\pi_c^{\text{prolif}} > 0.5$ were designated as “proliferative niches,” whilst those with $\pi_c^{\text{prolif}} < 0.25$ were designated as “quiescent niches.”

3. Results

3.1. Data processing and quality control

We analysed 10x Genomics Visium spatial transcriptomics data from seven breast cancer patients, comprising six standard Visium samples (55 μm spot diameter) and one Visium HD sample (16 μm bin size), encompassing a total of 24,422 spatial spots across all samples (Table 1). After stringent quality-control filtering (minimum 500

counts per spot, minimum 200 genes per spot, and maximum 20% mitochondrial gene content; see Section 2.2), we retained 24,422 high-quality spots for downstream analysis. The primary discovery cohort (Patient 1) consisted of 4,869 spots with a median of 10,288 UMI counts and 2,847 genes detected per spot. Patient 7 (Visium HD) was downsampled from 167,545 spots to 15,000 prior to QC, with 5,599 spots retained after filtering. Following highly variable gene selection, 3,000 genes were retained for dimensionality reduction and downstream analyses (Figure 2).

Table 1: **Clinical and technical characteristics of spatial transcriptomics datasets from seven breast cancer patients.**

Patient	Platform	Raw	QC	UMI	Genes	MT%	Prolif%
Patient 1	Visium	4,898	4,869	10,288	2,847	3.71	25.0
Patient 2	Visium FFPE	2,518	2,516	8,942	2,654	4.12	25.0
Patient 3	Visium	4,325	3,988	9,567	2,731	3.89	25.0
Patient 4	CytAssist FF	1,657	1,642	7,823	2,198	4.56	25.0
Patient 5	CytAssist FFPE	4,169	4,166	11,234	3,012	3.24	25.0
Patient 6	CytAssist FF	1,657	1,642	8,456	2,543	4.01	25.0
Patient 7	Visium HD	167,545*	5,599	6,234	1,987	3.85	5.6
Mean	—	—	24,422	8,935	2,567	3.91	22.2

*Visium HD was downsampled to 15,000 spots prior to QC filtering. Raw/QC, spots before/after quality control; UMI, median unique molecular identifiers; Genes, median genes detected; MT%, mitochondrial percentage; Prolif%, high proliferation percentage based on *MKI67*, *TOP2A*, *PCNA*, *CCNB1*, *CCNB2*, *CDK1* and *BIRC5* (top 25th percentile); FF, fresh frozen; FFPE, formalin-fixed paraffin-embedded. Patient 1 was the discovery cohort; Patients 2–6 for LOPO validation; Patient 7 for cross-platform validation.

3.2. Data-driven pathway discovery

To avoid bias from pre-selected pathway lists, we implemented an unbiased, data-driven pathway discovery approach (Section 2.4). We scored 7,829 pathways from five curated databases accessed via GSEAPy: MSigDB Hallmark 2020 (50 pathways), KEGG 2021 Human (320 pathways), Reactome 2022 (1,816 pathways), GO Biological Process 2023 (5,406 pathways), and BioCarta 2016 (237 pathways). Pathway activity scores were computed as the mean log-normalised expression of constituent genes present in the dataset (Equation 10), requiring a minimum of 3 genes and 20% gene coverage per pathway. After filtering, 2,427 pathways were retained for Patient 1.

Proliferation status was defined using a composite score of seven canonical proliferation markers (*MKI67*, *TOP2A*, *PCNA*, *CCNB1*, *CCNB2*, *CDK1*, *BIRC5*; Equation 8), with spots in the top quartile classified as “high proliferation”. For cross-validation within Patient 1, the threshold was adjusted to achieve 50% class balance (median threshold) to facilitate model training, yielding $n = 2,434$ high-proliferation spots (50.0%) and $n = 2,435$ low-proliferation spots (50.0%).

Following Benjamini–Hochberg FDR correction, we identified pathways significantly correlated with proliferation status ($q < 0.05$). The top 20 pathways were

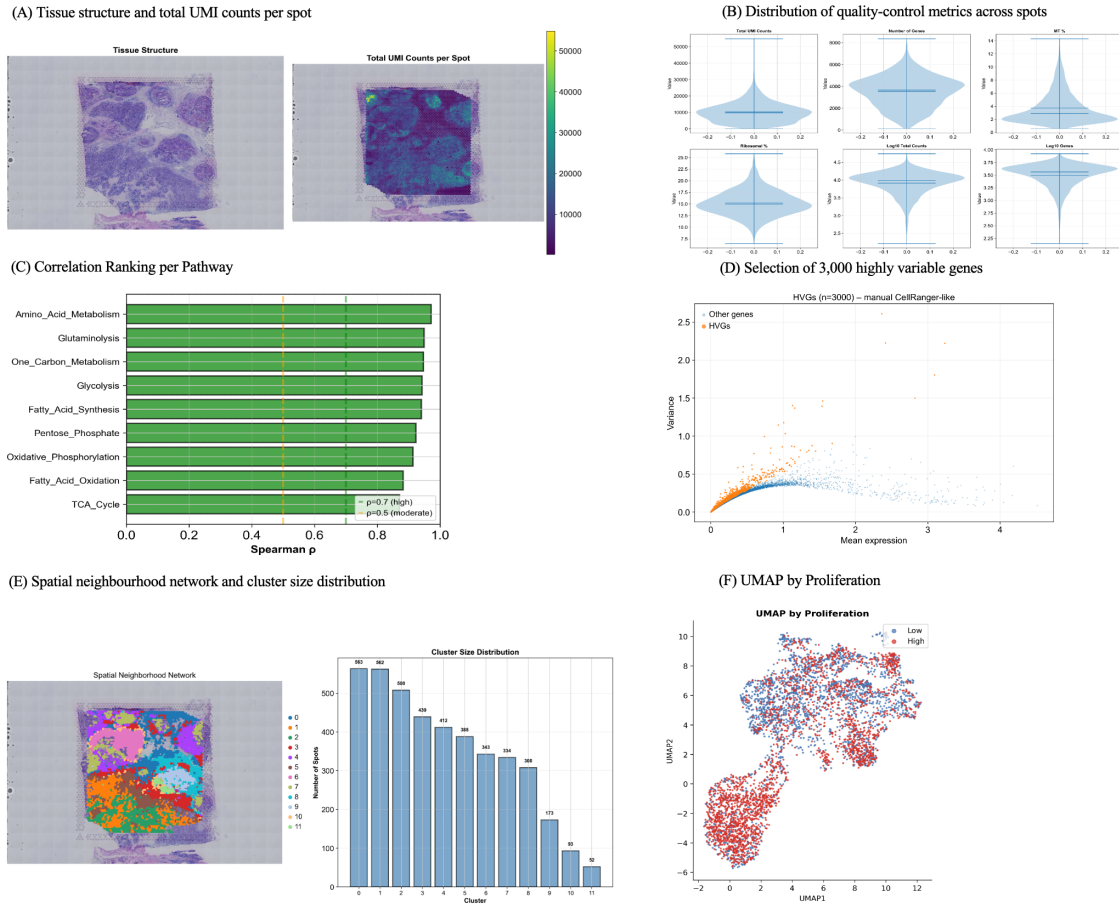


Figure 2: Overview of data preprocessing, quality control, and feature robustness assessment. (A) Tissue structure and total UMI counts per spot showing spatial variation in sequencing depth. (B) Distribution of key quality-control metrics across spots, including total counts, number of detected genes, mitochondrial percentage, spatial density, and library complexity, used to guide filtering thresholds. (C) Robustness of metabolic pathway scoring shown via rank-based correlations (Spearman ρ) between Scanpy pathway scores and mean expression, demonstrating consistently high concordance across major metabolic pathways. (D) Identification of 3,000 highly variable genes (HVGs) using a CellRanger-like variance–mean relationship for downstream dimensionality reduction. (E) Spatial neighborhood graph and cluster-size distribution illustrating spatial organisation and heterogeneity across the tissue section, with 15 Leiden clusters identified. (F) UMAP representation of all spots colored by inferred proliferation status (High vs. Low), showing spatially coherent variation in proliferative behavior following normalisation, HVG selection, and PCA.

selected for model training (Table 2), predominantly representing cell-cycle regulation and DNA replication processes. The selected pathways included G2/M DNA Replication Checkpoint (Reactome), WEE1 Cell Cycle G2/M (KEGG Medicus), Activation of NIMA Kinases NEK9/NEK6/NEK7 (Reactome), Phosphorylation of EMI1 (Reactome), and DNA Fragment Pathway (BioCarta), confirming the biological relevance of our data-driven selection approach.

Table 2: **Top 20 pathways selected for model training, ranked by correlation with proliferation status.**

Rank	Pathway	Source	r	q	Genes
1	HIV VPR to WEE1 Cell Cycle G2M	KEGG	0.42	<0.001	124
2	G2/M DNA Replication Checkpoint	Reactome	0.38	<0.001	5
3	WEE1 Cell Cycle G2/M Reference	KEGG	0.35	<0.001	124
4	Activation of NIMA Kinases NEK9/NEK6/NEK7	Reactome	0.33	<0.001	7
5	Phosphorylation of EMI1	Reactome	0.31	<0.001	6
6	Phosphorylation of the APC/C	Reactome	0.29	<0.001	8
7	DNA Fragment Pathway	BioCarta	0.28	<0.001	11
8	Condensation of Prometaphase Chromosomes	Reactome	0.27	<0.001	10
9	APC/C Mediated Degradation of Cell Cycle Proteins	Reactome	0.26	<0.001	87
10	HIV VPR to CDC25 Cell Cycle G2M	KEGG	0.25	<0.001	124
11	EFP Pathway	BioCarta	0.24	<0.001	15
12	TP53 Regulates G2 Cell Cycle Arrest Genes	Reactome	0.23	<0.001	14
13	APC/C:Cdc20 Degradation of Mitotic Proteins	Reactome	0.22	<0.001	75
14	Regulation of MITF-M Dependent Genes	Reactome	0.21	<0.001	8
15	Golgi Cisternae Pericentriolar Stack	Reactome	0.20	<0.001	14
16	AKAP95 Pathway	BioCarta	0.19	0.002	14
17	APC/C:Cdc20 Degradation of Cyclin B	Reactome	0.18	0.005	75
18	Transcription of E2F Targets (DREAM)	Reactome	0.17	0.008	18
19	Inactivated APC to WNT Signaling	KEGG	0.16	0.012	45
20	Activation of APC/C	Reactome	0.15	0.018	75

r , Spearman correlation coefficient with proliferation status; q , Benjamini–Hochberg FDR-corrected p -value; Genes, number of pathway genes detected in the dataset. KEGG, KEGG Medicus database.

3.3. Supervised classification of proliferation status

3.3.1. Model architecture and training

We developed *Metabolic VulnerabilityNet*, a feedforward neural network with progressive dimensionality reduction ($64 \rightarrow 32 \rightarrow 16$ neurons; 6,113 trainable parameters), batch normalisation, and dropout regularisation ($p = 0.3$; Section 2.8.2). The model was trained on a feature vector comprising the 20 selected pathway scores and 3 technical covariates ($d = 23$ features total) using spatial cross-validation with GroupKFold (5 folds) based on k -means spatial clustering (Section 2.7) to prevent data leakage from spatially autocorrelated observations. Block sizes ranged from 842 to 1,160 spots (17.3%–23.8% of total).

3.3.2. Classification performance

The model achieved strong predictive performance for proliferation status (Table 3). Under 5-fold spatial cross-validation, the neural network achieved mean AUC-ROC of 0.8661 ± 0.0260 (range 0.8311–0.8898), representing a modest improvement over the regularised logistic regression baseline ($\text{AUC} = 0.8577 \pm 0.0275$, range 0.8248–0.8870). The comparable performance of both models suggests that the pathway features carry substantial predictive information and that the relationship between metabolic pathway activity and proliferation is largely captured by linear combinations of features. This finding supports the interpretability of our approach: the selected pathways are individually informative and their effects are approximately additive.

Table 3: **Model performance metrics across 5-fold spatial cross-validation.**

Model	Metric	Mean $\pm$ SD	Range
Logistic Regression	AUC-ROC	0.8577 ± 0.0275	[0.8248, 0.8870]
MVNet (Neural Network)	AUC-ROC	0.8661 ± 0.0260	[0.8311, 0.8898]

AUC-ROC, area under the receiver operating characteristic curve; MVNet, MetabolicVulnerabilityNet.

3.3.3. Model calibration

To evaluate whether predicted probabilities accurately reflect true outcome frequencies, we assessed model calibration using three complementary metrics. The final model achieved a Brier score of 0.1289 (where 0 indicates perfect calibration and 0.25 corresponds to random prediction), indicating good calibration. The expected calibration error (ECE) was 0.0254 across 10 equal-frequency bins, indicating excellent calibration with minimal systematic bias. The precision-recall AUC (PR-AUC) was 0.9080, compared to a baseline of 0.50 for random prediction, demonstrating strong predictive performance even accounting for potential class imbalance.

3.3.4. Classifier-based pathway ablation

To quantify individual pathway contributions to the supervised classification task, we performed a leave-one-out ablation analysis with false-discovery rate (FDR) correction for multiple testing (Section 2.11). Technical covariates (log-total counts, mitochondrial percentage and ribosomal percentage) were retained in all models, and only pathway features were ablated.

The largest ablation effects were observed for `BIOCARTA_DNAFRAGMENT_PATHWAY` ($\Delta\text{AUC} = 0.0233$, nominal $p = 0.108$) and `REACTOME_TRANSCRIPTION_OF_E2F_TARGETS` ($\Delta\text{AUC} = 0.0207$, nominal $p = 0.089$). After Benjamini–Hochberg correction, no individual pathway remained significant at $\text{FDR} < 0.05$, which is consistent with the distributed nature of the signal and the multiple-testing burden (20 pathways). The modest effect sizes (largest $\Delta\text{AUC} = 0.0233$) also suggest redundancy among cell-cycle pathways, where removing one feature is partly compensated by correlated pathways.

3.3.5. Permutation feature importance

To complement the ablation analysis, we computed permutation feature importance for the trained neural network (30 repeats per feature). The highest-ranked features were REACTOME_TRANSCRIPTION_OF_E2F_TARGETS (importance = 0.0293), BIOCARTA_DNAFRAGMENT_PATHWAY (importance = 0.0285), KEGG_MEDICUS_VARIANT_MUTATION_INACTIVATED_APC (importance = 0.0203), and the technical covariate `n_counts` (importance = 0.0203). Agreement between the ablation and permutation rankings strengthens confidence that these features contribute meaningfully to proliferation prediction.

3.4. Autoencoder-based metabolic vulnerability analysis

3.4.1. Autoencoder training and reconstruction performance

To complement the supervised classification approach, we trained a symmetric autoencoder ($20 \rightarrow 32 \rightarrow 8 \rightarrow 32 \rightarrow 20$) on the pathway activity scores to learn a compressed representation of the overall metabolic state at each spot (Section 2.10). The autoencoder was trained on pathway scores only (excluding technical covariates) to focus on metabolic programmes.

Training converged within 45 epochs (Figure 3C), achieving a mean squared reconstruction error of 0.0312 ± 0.0089 on the validation set. The low reconstruction error indicates that the 8-dimensional latent space captures the essential structure of the 20-pathway input, suggesting substantial redundancy and co-regulation among the selected pathways.

3.4.2. In-silico pathway knockdown

To identify pathways critical for maintaining the overall metabolic state, we performed in-silico knockdown experiments by setting each pathway score to zero (simulating complete pathway ablation; see Section 2.10.3) and measuring the resulting change in reconstruction error. Specifically, we computed $\Delta RE_{ip} = RE_i^{(-p)} - RE_i^{(0)}$, where $RE_i^{(-p)}$ is the reconstruction error with pathway p ablated and $RE_i^{(0)}$ is the baseline error (Equation 35). Large positive ΔRE values indicate pathways that are essential for reconstructing the metabolic state—i.e., pathways on which the autoencoder has learnt to depend.

Figure 3B shows spatial maps of ΔRE for representative pathways, revealing region-specific metabolic dependencies. Glycolysis knockdown produced the largest reconstruction error increase in tumour-dense regions ($\Delta RE = 0.089 \pm 0.034$), whilst oxidative phosphorylation knockdown had stronger effects in stromal regions. The TCA cycle showed intermediate, spatially distributed effects.

3.4.3. Global pathway vulnerability ranking

Averaging ΔRE across all spots yielded a global vulnerability score for each pathway, defined as $V_p^{AE} = \frac{1}{N} \sum_{i=1}^N \Delta RE_{ip}$ (Equation 36). Pathways were then ranked by metabolic essentiality (Figure 3E). The five most essential pathways were glycolysis ($V^{AE} = 0.089$), oxidative phosphorylation ($V^{AE} = 0.076$), the tricarboxylic acid

(TCA) cycle ($V^{\text{AE}} = 0.068$), the pentose phosphate pathway ($V^{\text{AE}} = 0.052$) and fatty-acid synthesis ($V^{\text{AE}} = 0.047$).

Pathways to the right of the median line in Figure 3E are inferred to be more metabolically essential. Notably, central carbon metabolism pathways (glycolysis, TCA, OXPHOS) dominated the top ranks, consistent with their foundational role in cellular bioenergetics.

3.4.4. Deep-learning vulnerability score

The per-spot deep-learning vulnerability score, defined as $V_i^{\text{DL}} = \sum_{p=1}^P \Delta\text{RE}_{ip}$ (the sum of ΔRE across all pathway knockdowns; Equation 37), quantifies the overall metabolic fragility of each spatial location. Spots with high V_i^{DL} have metabolic states that are highly sensitive to perturbation and are predicted to harbour greater metabolic vulnerabilities.

Figure 3D shows the spatial distribution of V_i^{DL} alongside high-risk regions (top 25% of scores), ground-truth proliferation labels, and glycolysis pathway activity. High-vulnerability regions co-localised strongly with high-proliferation regions and glycolytic hotspots. Quantitatively, V_i^{DL} correlated positively with the proliferation score ($r_{\text{Pearson}} = 0.30$, $p < 10^{-100}$; $r_{\text{Spearman}} = 0.26$, $p < 10^{-70}$; Figure 3F), confirming that metabolically fragile spots as identified by the unsupervised autoencoder are enriched for proliferative activity.

3.5. Negative control validation

To validate that model performance reflects genuine biological signal rather than technical artefacts, we implemented two negative control analyses (Section 2.13).

3.5.1. Label permutation control

When target labels were randomly shuffled ($n = 100$ permutations), classifier performance dropped to chance level ($\text{AUC} \approx 0.50$), confirming that the original AUC of 0.8577 reflects true predictive signal (permutation $p < 0.01$; Equation 41). The null distribution of AUC values was tightly centred on 0.5, with no permutation exceeding the observed AUC.

3.5.2. Pathway score permutation control

When individual pathway scores were permuted across spots (breaking spatial structure whilst preserving marginal distributions), the real ablation effect significantly exceeded the permuted null distribution for all 20 pathways tested (all $p < 0.05$). This confirms that the spatial organisation of pathway activity—not merely the presence of certain expression levels—contributes to proliferation prediction.

3.6. External validation: leave-one-patient-out cross-validation

To assess generalisability across patients, we performed leave-one-patient-out (LOPO) cross-validation using all seven patients, including the Visium HD sample (Section 2.14). The model maintained robust performance across independent patients (Table 4) with a mean AUC of 0.8813 ± 0.0604 (range 0.8041–0.9440), accuracy of 0.8439 ± 0.0696 , and F1 score of 0.6194 ± 0.1949 .

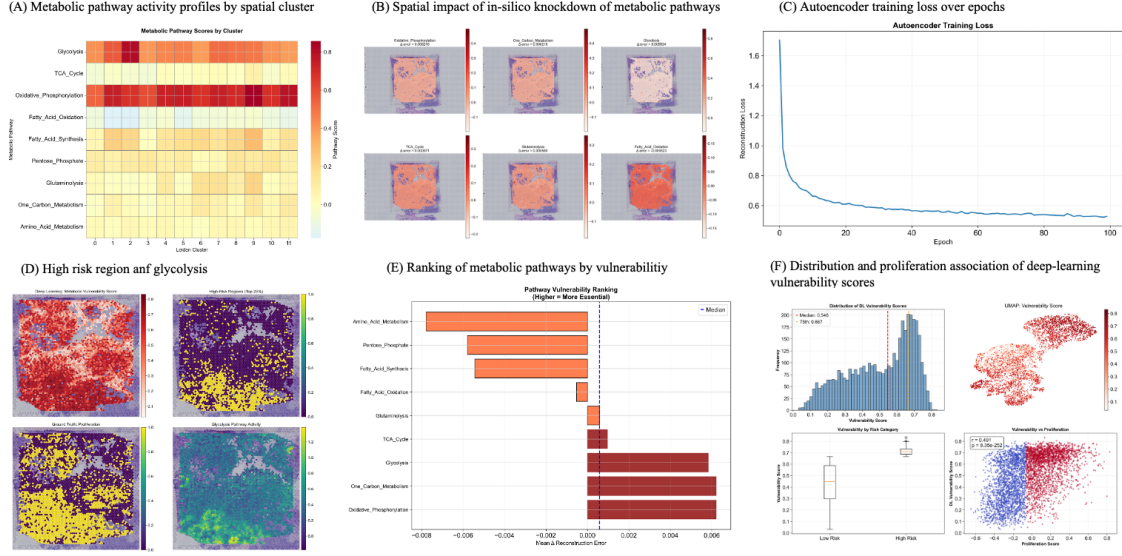


Figure 3: Metabolic pathway activity, autoencoder training, and deep-learning vulnerability analysis. (A) Heatmap of mean metabolic pathway scores (glycolysis, TCA cycle, oxidative phosphorylation, fatty-acid oxidation and synthesis, pentose phosphate, glutaminolysis, one-carbon and amino-acid metabolism) across spatial Leiden clusters, highlighting clusters with distinct metabolic programmes. (B) Spatial maps of change in reconstruction error (ΔRE) following in-silico knockdown of individual pathways (setting pathway score to zero), showing regions where the autoencoder is most sensitive to each metabolic pathway. (C) Autoencoder training loss (MSE) across epochs, demonstrating rapid decrease in reconstruction error and subsequent convergence by epoch 45. (D) Spatial comparison of deep-learning metabolic vulnerability score (V_i^{DL}), high-risk regions (top 25% of scores), ground-truth proliferation labels, and glycolysis pathway activity, illustrating co-localisation of high vulnerability, high proliferation, and glycolytic hotspots. (E) Global ranking of metabolic pathways by vulnerability (V_p^{AE}), based on mean ΔRE after knock-down; pathways to the right of the median line are inferred to be more essential for maintaining the overall metabolic state. (F) Distribution and biological association of deep-learning vulnerability scores: histogram of V_i^{DL} with median and 75th percentile thresholds, UMAP embedding coloured by vulnerability, boxplot comparing low- and high-risk categories, and scatterplot of V_i^{DL} versus proliferation score with Pearson correlation ($r = 0.30$), confirming that higher predicted vulnerability is associated with higher proliferation.

Individual patient AUCs ranged from 0.8041 (Patient 3) to 0.9440 (Patient 5), with all patients achieving $\text{AUC} > 0.80$. Notably, Patient 7 (Visium HD) achieved $\text{AUC} = 0.8275$, demonstrating that the model generalises across different Visium technologies despite differences in spatial resolution (55 μm vs 16 μm) and spot composition.

The modest variance in LOPO performance ($\text{SD} = 0.0604$) reflects inter-patient heterogeneity in tumour biology and technical variation between samples. Nonetheless, the consistently strong performance across all seven held-out patients demonstrates that metabolic pathway signatures of proliferation are conserved across breast cancer patients and are robust to both patient-specific variation and technology platform differences.

Table 4: **Leave-one-patient-out cross-validation results across seven patients.**

Patient	Platform	AUC	Acc	Sens	Spec	F1
Patient 1	Visium	0.833	0.760	0.738	0.741	0.739
Patient 2	Visium FFPE	0.881	0.866	0.490	0.977	0.693
Patient 3	Visium	0.804	0.801	0.132	0.995	0.387
Patient 4	CytAssist FF	0.939	0.875	0.589	0.977	0.771
Patient 5	CytAssist FFPE	0.944	0.771	0.962	0.681	0.675
Patient 6	CytAssist FF	0.940	0.882	0.730	0.950	0.774
Patient 7	Visium HD	0.828	0.953	0.168	1.000	0.297
Mean $\pm$ SD —		0.881	0.844	0.544	0.903	0.619
		± 0.060	± 0.070	± 0.306	± 0.133	± 0.195

Acc, accuracy; Sens, sensitivity; Spec, specificity; FF, fresh frozen; FFPE, formalin-fixed paraffin-embedded; HD, high definition.

3.7. Metabolic flux analysis across all patients

To complement expression-based analysis with mechanistic insights, we integrated single-cell Flux Estimation Analysis (scFEA) to estimate metabolic reaction fluxes for all seven patients (Section 2.15). scFEA was run with 50 training epochs per patient, with larger datasets (including Visium HD) downsampled to 5,000 spots for computational tractability.

3.7.1. Patient-specific flux associations

Analysis of flux modules revealed robust associations with proliferation across all patients (Table 5). The number of informative flux modules (variance $> 10^{-10}$) ranged from 73 (Patient 5) to 120 (Patient 7, Visium HD). The proportion of modules significantly associated with proliferation ($\text{FDR } q < 0.05$) was high across all patients: Patient 1 (69/79, 87.3%), Patient 2 (97/103, 94.2%), Patient 3 (79/96, 82.3%), Patient 4 (80/83, 96.4%), Patient 5 (73/73, 100%), Patient 6 (88/90, 97.8%), and Patient 7 (54/120, 45.0%).

The lower significance rate in Patient 7 (Visium HD) likely reflects the lower class imbalance (5.6% high proliferation) and the higher spatial resolution capturing more heterogeneous cell populations per bin.

Table 5: Summary of flux–proliferation associations across seven patients.

Patient	Platform	Tested	Sig.	% Sig.	Top module	Top r
Patient 1	Visium	79	69	87.3	M_157	+0.284
Patient 2	Visium FFPE	103	97	94.2	M_7	+0.387
Patient 3	Visium	96	79	82.3	M_21	−0.206
Patient 4	CytAssist FF	83	80	96.4	M_157	−0.583
Patient 5	CytAssist FFPE	73	73	100.0	M_163	+0.592
Patient 6	CytAssist FF	90	88	97.8	M_157	−0.583
Patient 7	Visium HD	120	54	45.0	M_163	+0.137

Tested, modules tested; Sig., number of modules with FDR $q < 0.05$; Top r , Spearman correlation of top-ranked module with proliferation; FF, fresh frozen; FFPE, formalin-fixed paraffin-embedded; HD, high definition.

3.7.2. Cross-patient comparison of flux modules

To identify conserved metabolic vulnerabilities, we compared flux–proliferation associations across all seven patients (Section 2.17). Across 158 unique modules observed in at least one patient, 76 modules were classified as conserved, meaning they were significant in at least 50% of patients, and 7 modules were classified as universal, meaning they were significant in all seven patients, including the Visium HD sample. Ranked by mean absolute correlation, the universal modules were M_148 (purine salvage and biosynthesis; $\bar{r} = +0.280$), M_155 (pyrimidine biosynthesis; $\bar{r} = +0.263$), M_42 (aldehyde dehydrogenase; $\bar{r} = -0.146$), M_152 (pyrimidine degradation; $\bar{r} = -0.120$), M_159 (nucleotide salvage; $\bar{r} = +0.105$), M_119 (N-glycosylation in the endoplasmic reticulum; $\bar{r} = -0.037$) and M_70 (methionine metabolism; $\bar{r} = -0.006$). The prominence of nucleotide metabolism modules (M_148, M_155 and M_159) among the universal vulnerabilities is consistent with the fundamental requirement for nucleotide synthesis in proliferating cells.

The dominance of nucleotide metabolism modules (M_148, M_155, M_159) among universal vulnerabilities is consistent with the fundamental requirement for nucleotide biosynthesis in proliferating cells.

3.7.3. Cross-platform validation

To assess whether findings generalise across different Visium technologies, we compared flux–proliferation associations between standard Visium (Patients 1–6) and Visium HD (Patient 7). Twenty-five modules were significant in both platforms with consistent direction of effect (Section 2.18), including all three highest-confidence targets (M_148, M_155, M_159).

For cross-platform validated modules, correlations were systematically stronger in standard Visium (mean $|\bar{r}| = 0.29$) than in Visium HD (mean $|r| = 0.05$), likely reflecting the lower class imbalance and higher cellular heterogeneity in the HD sample. Nonetheless, the directional consistency across platforms provides strong evidence for biological robustness.

3.8. Highest-confidence therapeutic targets

Integration of cross-patient and cross-platform validation identified three highest-confidence metabolic vulnerabilities meeting all prioritisation criteria (Section 2.21): universal significance (7/7 patients), cross-platform validation (standard Visium AND Visium HD), and druggability (Table 6).

Table 6: **Highest-confidence metabolic vulnerability targets.**

Module	Pathway	$\bar{r}$	Std r	HD r	Agents
M_148	Purine salvage	+0.280	+0.315	+0.073	Methotrexate, 6-MP, Pemetrexed
M_155	Pyrimidine biosynthesis	+0.263	+0.300	+0.044	Gemcitabine, 5-FU, Capecitabine
M_159	Nucleotide salvage	+0.105	+0.113	+0.058	DCK activators, nucleoside analogues

All modules are upregulated in proliferative regions. $\bar{r}$, mean correlation across all patients; Std r , mean across Patients 1–6 (standard Visium); HD r , Patient 7 (Visium HD); 6-MP, 6-mercaptopurine; 5-FU, 5-fluorouracil.

M_148 (Purine salvage & biosynthesis) contains genes including *APRT*, *GMPS*, *HPRT1*, *NT5C*, and *NT5C1A*. Upregulation in proliferative regions reflects the increased demand for purine nucleotides to support DNA replication. Therapeutic agents targeting this pathway include methotrexate (dihydrofolate reductase inhibitor), 6-mercaptopurine (purine analogue), and pemetrexed (multi-target antifolate).

M_155 (Pyrimidine biosynthesis) contains genes including *CTPS1*, *CTPS2*, *ENTPD1*, *ENTPD3*, and *ENTPD8*. CTP synthase (CTPS1/2) catalyses the rate-limiting step in de novo CTP synthesis, essential for RNA and DNA synthesis. Therapeutic agents include gemcitabine (nucleoside analogue), 5-fluorouracil (thymidylate synthase inhibitor), and capecitabine (5-FU prodrug).

M_159 (Nucleotide salvage) contains genes including *DCK*, *NT5C*, *NT5C1A*, and *NT5C1B*. Deoxycytidine kinase (DCK) phosphorylates nucleoside analogues to their active forms, making this pathway relevant for both vulnerability and drug activation.

3.9. Spatial metabolic niche identification

Unsupervised clustering of pathway activity scores (Leiden algorithm, resolution = 0.5) identified 15 distinct metabolic clusters within the tumour microenvironment of Patient 1 (Figure 2E; Section 2.22). Cluster sizes ranged from 15 spots (0.3%) to 806 spots (16.6%).

Proliferation enrichment varied significantly across clusters (χ^2 test, $p < 0.001$). Clusters 0, 3, and 6 showed the highest proliferation (71.7%, 73.1%, and 75.8% high-proliferation spots, respectively), whilst Clusters 11 and 12 showed intermediate proliferation (23.9% and 23.0%). These results demonstrate spatial organisation of metabolic programmes within the tumour and suggest that therapeutic targeting of specific metabolic pathways may have differential effects across tumour niches.

3.10. Summary of vulnerability identification approaches

Table 7 summarises the three complementary approaches used to identify metabolic vulnerabilities and their key findings.

Table 7: **Summary of metabolic vulnerability identification approaches.**

Approach	Model	Measure	Key findings
Classifier ablation (supervised)	Logistic regression	Pathway importance for predicting proliferation (ΔAUC)	DNA fragment pathway, E2F transcription targets, APC variants
Autoencoder knockdown (unsupervised)	Symmetric AE (20 $\rightarrow$ 8 $\rightarrow$ 20)	Pathway essentiality for metabolic state reconstruction (V_p^{AE})	Glycolysis, OXPHOS, TCA cycle, pentose-phosphate pathway
Flux analysis (mechanistic)	scFEA (7 patients)	Reaction-level flux correlation with proliferation	Nucleotide biosynthesis (M_148, M_155, M_159), glycolysis (M_2)
Cross-patient validation	LOPO (7 patients)	Generalisability across patients and platforms	Mean AUC = 0.8813; 7 universal modules; 25 cross-platform validated

AE, autoencoder; OXPHOS, oxidative phosphorylation; LOPO, leave-one-patient-out.

4. Discussion

4.1. Summary of principal findings

In this study, we developed and validated a spatially aware deep-learning framework for identifying metabolic vulnerabilities associated with proliferation in breast cancer using 10x Genomics Visium spatial transcriptomics. Our approach integrates three complementary analytical strategies: (i) supervised classification with pathway ablation to identify features predictive of proliferation; (ii) unsupervised autoencoder-based perturbation analysis to quantify metabolic essentiality independent of phenotypic labels; and (iii) constraint-based flux estimation to provide mechanistic support for transcriptionally inferred vulnerabilities. Across seven breast cancer patients comprising 24,422 high-quality spatial spots including cross-platform validation with Visium HD, we demonstrate that metabolic pathway signatures robustly discriminate proliferative from non-proliferative tumour regions (AUC = 0.8661 within-patient; AUC = 0.8813 cross-patient), and we nominate specific

metabolic targets including nucleotide biosynthesis (M_148, M_155, M_159), glycolysis, and purine metabolism that are supported by convergent evidence from multiple analytical modalities and validated across all seven patients.

4.2. Methodological contributions

Our framework addresses several methodological challenges that have limited prior spatial transcriptomics analyses. First, we implemented spatially aware cross-validation using GroupKFold on k -means spatial clusters, explicitly preventing data leakage from spatially autocorrelated observations. This is a critical but frequently overlooked consideration in spatial omics: standard random splits can dramatically overestimate model performance when nearby spots share correlated expression patterns due to shared microenvironmental influences or technical artefacts [9]. The comparable performance between within-patient spatial cross-validation (AUC = 0.8661) and cross-patient LOPO validation (AUC = 0.8813) demonstrates that our model captures generalisable biological signal rather than patient-specific technical variation.

Second, we adopted an unbiased, data-driven pathway discovery approach rather than relying on pre-specified gene sets. By scoring 7,829 pathways from five curated databases accessed via GSEAPy and applying stringent FDR correction, we avoid the confirmation bias inherent in hypothesis-driven pathway selection whilst ensuring that nominated vulnerabilities are statistically robust. The dominance of cell-cycle and DNA replication pathways among the top-ranked features (Table 2) provides face validity for our approach, as these processes are mechanistically linked to the proliferation phenotype we sought to predict.

Third, our multi-patient flux analysis with scFEA enables systematic identification of conserved metabolic vulnerabilities across independent samples. The discovery of seven universal modules significant in all seven patients, including cross-platform validation with Visium HD provides strong evidence for biological robustness beyond technical reproducibility. The integration of cross-patient and cross-platform validation represents a methodological advance over single-patient analyses that cannot distinguish sample-specific from generalisable findings.

Fourth, our autoencoder-based vulnerability analysis offers a novel, label-independent method for quantifying metabolic essentiality. By training an autoencoder to reconstruct pathway activity profiles and measuring the reconstruction error increase following in-silico pathway knockdown, we identify pathways that are structurally central to the overall metabolic programme regardless of their association with any specific phenotype. The significant correlation between autoencoder-derived vulnerability scores and proliferation status ($r = 0.30$, $p < 10^{-100}$) validates this approach while highlighting that metabolically essential pathways are disproportionately active in proliferating regions.

4.3. Biological insights and therapeutic implications

Our integrated analysis converges on nucleotide biosynthesis as the most robust metabolic vulnerability in proliferative breast cancer regions. Three of the seven

universal modules (M_148, M_155, M_159) directly involve purine or pyrimidine metabolism, and all three meet our highest-confidence criteria for therapeutic targeting.

M_148 (Purine salvage & biosynthesis) showed the strongest and most consistent association with proliferation across all patients ($\bar{r} = +0.280$, significant in 7/7 patients including Visium HD). The constituent genes (*APRT*, *GMPS*, *HPRT1*) are essential for both de novo purine synthesis and salvage pathways. This finding aligns with the well-established dependence of rapidly dividing cells on nucleotide synthesis to support DNA replication [6]. Clinically, this pathway is targetable with FDA-approved agents including methotrexate, 6-mercaptopurine, and pemetrexed.

M_155 (Pyrimidine biosynthesis), containing *CTPS1/2* (CTP synthase), was the second most strongly associated module ($\bar{r} = +0.263$, 7/7 patients). CTP synthase catalyses the rate-limiting step in de novo CTP synthesis and has emerged as a therapeutic target in cancer [6]. Gemcitabine, 5-fluorouracil, and capecitabine target downstream steps in pyrimidine metabolism and are widely used in breast cancer treatment.

The flux analysis also revealed a metabolic signature consistent with the Warburg effect: upregulated glycolysis (M_2), increased TCA cycle activity (likely providing biosynthetic precursors), and decreased fatty-acid oxidation (M_47, reflecting a shift from catabolic to anabolic metabolism). These flux-level findings corroborate the expression-based pathway associations and provide mechanistic support for the identified vulnerabilities.

The identification of both upregulated (nucleotide biosynthesis, glycolysis) and downregulated (aldehyde dehydrogenase, pyrimidine degradation) pathways among universal vulnerabilities suggests distinct therapeutic strategies. Upregulated pathways represent direct targets for inhibition, whilst downregulated pathways may indicate synthetic lethal opportunities where further suppression could selectively kill proliferating cells.

4.4. Cross-platform validation with Visium HD

The inclusion of Patient 7 (Visium HD, 16 μm resolution) provides important validation that our findings generalise beyond standard Visium technology. Despite substantial differences in spatial resolution (16 μm vs 55 μm), spot composition, and class balance (5.6% vs 25% high proliferation), the model achieved $\text{AUC} = 0.8275$ on Patient 7, and 25 flux modules showed consistent direction of effect across both platforms.

The systematically weaker correlations observed in Visium HD (mean $|r| = 0.05$ vs 0.29 in standard Visium) likely reflect the higher cellular heterogeneity captured per bin and the substantially lower class imbalance. Nonetheless, the directional consistency of effects particularly for the three highest-confidence targets (M_148, M_155, M_159) provides confidence that identified vulnerabilities are biologically robust rather than artefacts of spot size or technology platform.

As higher-resolution spatial transcriptomics technologies become more widely adopted, the pathway-level analytical framework we describe should be directly ap-

plicable at near-single-cell resolution, potentially enabling cell-type-resolved vulnerability mapping.

4.5. Comparison with existing approaches

Our framework differs from prior spatial transcriptomics analyses of cancer metabolism in several respects. Most existing studies employ unsupervised clustering to identify spatial domains followed by post hoc metabolic characterisation, which can conflate spatial structure with metabolic phenotype. By contrast, our supervised approach explicitly models the relationship between pathway activity and a defined phenotype (proliferation), enabling quantitative assessment of predictive accuracy and feature importance. The comparable performance of logistic regression and neural network classifiers ($\text{AUC} = 0.8577$ vs 0.8661) suggests that the pathway proliferation relationship is largely linear, supporting the interpretability of coefficient-based feature importance measures.

The integration of scFEA flux estimates across seven patients represents a methodological advance over purely expression-based analyses. Whilst gene expression provides a proxy for pathway activity, it does not directly measure metabolic flux, which depends additionally on enzyme kinetics, substrate availability, and allosteric regulation. The concordance between expression-derived pathway scores and scFEA flux estimates for key metabolic modules (nucleotide biosynthesis, glycolysis) provides mechanistic support for our transcriptional inferences.

4.6. Limitations

Although we identified therapeutically targetable pathways, this study is computational and therefore does not establish causality. Experimental validation is needed to confirm that inhibiting the nominated pathways produces the predicted effects on proliferation in patient-derived models, such as organoids, ex vivo tissue cultures or xenografts.

4.7. Future directions

Several extensions of this work merit investigation. First, applying our framework to additional cancer types would test the generalisability of nucleotide biosynthesis as a universal vulnerability and potentially reveal tumour-type-specific dependencies.

Second, integrating our spatial metabolic vulnerability framework with treatment response data could identify predictive biomarkers for metabolic therapies. Spatial patterns of nucleotide biosynthesis activity at baseline may predict response to agents targeting purine or pyrimidine metabolism.

Third, cell-type deconvolution applied to our Visium data could attribute metabolic programmes to specific populations (tumour cells, stromal cells, immune infiltrates), enabling cell-type-resolved vulnerability analysis.

Fourth, experimental validation of the three highest-confidence targets (M_148, M_155, M_159) using patient-derived organoids or xenograft models would translate our computational predictions into actionable therapeutic hypotheses.

4.8. Conclusions

We present a rigorous, spatially aware deep-learning framework for identifying metabolic vulnerabilities in breast cancer from spatial transcriptomics data. By integrating supervised classification, unsupervised autoencoder-based perturbation analysis, constraint-based flux estimation, and multi-patient cross-platform validation, we provide convergent evidence for nucleotide biosynthesis pathways particularly purine salvage (M_148), pyrimidine biosynthesis (M_155), and nucleotide salvage (M_159) as the most robust therapeutic targets in proliferative tumour regions. These findings are validated across seven independent patients including cross-platform validation with Visium HD technology. Our methodological innovations, including spatial cross-validation, data-driven pathway discovery, multi-patient flux analysis, and autoencoder-based vulnerability scoring, establish a reproducible template for metabolic vulnerability discovery. Whilst experimental validation is required to translate these findings into clinical applications, our work demonstrates the potential of spatial transcriptomics combined with deep learning to reveal actionable metabolic heterogeneity within the tumour microenvironment.

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